

Repairing the linearization radius in shadow tomography

Yuxuan Zhang

École Polytechnique Fédérale de Lausanne (EPFL), 1015 Lausanne, Switzerland
Department of Physics, Princeton University, Princeton, NJ 08544, USA

zhangyx@comp.nus.edu.sg

17 June 2026*

Abstract

The linearization argument behind the shadow-tomography bounds of Chen, Li and Liu is valid only inside a radius $\|E\|_F \leq (0.01/t)^4$. We record two things: the underlying tail lemma does not close as printed, the $j = 2$ term exceeding the stated bound by a factor of order 10^{11} independently of the radius; and the constant in the Haar-moment estimate is slack enough that the radius can be replaced by $\|E\|_F \leq 1/(4et)$ — fourth power to first. Downstream the precision exponent moves from 12 to $5/2$ for balanced and structured observables, and to $11/2$ in general, with copy count unchanged. We note that the framework itself was subsequently bypassed.

1. The radius

The analysis of [1] linearizes around the maximally mixed state, and the linearization is controlled only for $\|E\|_F \leq (0.01/t)^4$, with the moment lemma proved in [2]. That fourth power is what confines the whole argument to very high precision.

2. The lemma does not close as printed

The tail sum $\sum_{j \geq 2} 2^j \binom{t}{j}^2 (8j)^{8j} \|E\|^{2j}$ is asserted to be at most $(100t)^4 \|E\|^4$. The $j = 2$ term alone exceeds this by a factor of order 1.8×10^{11} , and the excess does not depend on $\|E\|$, so shrinking the radius does not rescue it. The replacement below therefore repairs the lemma rather than sharpening it.

3. The constant is slack

Using the exact identity

$$\|\mathbb{E}_U[(UEU^\dagger)^{\otimes t'}]\|_{\text{op}} = \max_{\lambda} \frac{|s_{\lambda}(\alpha)|}{\dim V_{\lambda}^{t'}},$$

together with $|s_{\lambda}| \leq N_1(t') f^{\lambda}$ and $\max R = t^{t'}/t'! \leq e^{t'}$, the per-term constant is bounded by $t^{t'}$. Re-running the tail sum gives

$$\|E\|_F \leq \frac{1}{4et}$$

in place of $(0.01/t)^4$, with a better constant besides. Downstream the precision exponent falls from 12 to $5/2$ for balanced and structured observables and to $11/2$ in general, with copy complexity unchanged at $O(\log(1/\delta)/\varepsilon^2)$.

*Incoming: Department of Computer Science and Department of Physics, National University of Singapore, December 2026.

4. Verification, and an exposition slip

The Murnaghan–Nakayama table was rebuilt from scratch and cross-checked against the power-sum and bialternant formulas; the logical link was confirmed on 40k signed traceless spectra with no violation; the tail ratio stays below 0.78 uniformly to $t = 5000$. Two accounting gaps found adversarially were closed, both exponent-neutral. One is an exposition slip in [1], where a median bound is used as though it were a mean; the quantity is recoverable from an exact Schur–Weyl second moment, so nothing downstream breaks.

5. Placement

All of the above lives inside the linearization framework, which was subsequently bypassed: [3] gives an exactly unbiased estimator, which has no bias radius to respect and reaches $\varepsilon \lesssim d^{-1}$ directly. This note is therefore of interest for the technique and for the record on the lemma, not for the state of the art on the problem.

Disclosure

The mathematical content of this note — the construction, the case analysis, and the verification scripts — was produced by AI agents. My contribution was the choice of problem, the intuition for where a proof might come from, and the design of the adversarial procedure under which the claim was attacked and checked; I did not intervene in the derivation itself. Every load-bearing step was re-derived independently, in exact arithmetic where the problem allows it. This note is a record of a computer-assisted result. It has not been refereed.

References

- [1] S. Chen, J. Li and A. Liu, *Optimal high-precision shadow tomography*, arXiv:2407.13874.
- [2] S. Chen, J. Li and A. Liu, *An optimal tradeoff between entanglement and copy complexity for state tomography*, arXiv:2402.16353.
- [3] A. Pelecanos, X. Spilecki and J. Wright, *The debiased Keyl’s algorithm*, arXiv:2510.07788.